CSE STUDY HUB

Generative AI & Prompt Engineering

UNIT 5 · AI Intervention

What this unit is about

This unit builds understanding of **AI Intervention**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. AI-driven automation and decision-support systems

2. Human-AI collaboration and human-in-the-loop review

3. Bias, fairness, transparency and accountability

4. Privacy, consent, data security and intellectual property

5. Safety: harmful outputs, misuse, robustness and guardrails

6. Responsible AI lifecycle: assess, monitor, audit and improve

7. Measuring intervention impact: quality, time, cost and user outcomes

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 5 NOTES · PART 1

AI-driven automation and decision-support systems

AI-driven automation and decision-support systems is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Human-AI collaboration and human-in-the-loop review

Human-AI collaboration and human-in-the-loop review is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Bias, fairness, transparency and accountability

Bias, fairness, transparency and accountability is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on AI Intervention, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 2

Privacy, consent, data security and intellectual property

For Privacy, consent, data security and intellectual property, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Safety: harmful outputs, misuse, robustness and guardrails

Safety: harmful outputs, misuse, robustness and guardrails is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on AI Intervention, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 5 NOTES · PART 3

Responsible AI lifecycle: assess, monitor, audit and improve

Present Responsible AI lifecycle: assess, monitor, audit and improve as an ordered workflow. State the preconditions, list the major steps, explain what each step produces, and mention how errors are detected or handled. Use a small worked example or flowchart in the exam.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Measuring intervention impact: quality, time, cost and user outcomes

Measuring intervention impact: quality, time, cost and user outcomes is a syllabus concept in Generative AI & Prompt Engineering. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on AI Intervention, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **AI-driven automation and decision-support systems** — know the definition, core working and one example.

• **Human-AI collaboration and human-in-the-loop review** — know the definition, core working and one example.

• **Bias, fairness, transparency and accountability** — know the definition, core working and one example.

• **Privacy, consent, data security and intellectual property** — know the definition, core working and one example.

• **Safety: harmful outputs, misuse, robustness and guardrails** — know the definition, core working and one example.

• **Responsible AI lifecycle: assess, monitor, audit and improve** — know the definition, core working and one example.

• **Measuring intervention impact: quality, time, cost and user outcomes** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from AI Intervention. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define AI-driven automation and decision-support systems and explain its importance.

Q2. Explain the working/steps involved in Human-AI collaboration and human-in-the-loop review.

Q3. Differentiate between two related concepts from AI Intervention.

Q4. Give one practical application of Measuring intervention impact: quality, time, cost and user outcomes and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of AI Intervention.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.